
Understanding Linear Regions of the ReLU Networks

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Abstract

We focus on understanding the linear regions of ReLU (Rectified Linear Unit) networks. We aim to explore the behavior of ReLU networks with a particular focus on their linear regions. The article summarizes the general structure and key points discussed in each chapter. We explain how ReLU networks can be divided into linear regions where the function is linear and non-linear regions where the function is nonlinear. Additionally, we discuss how linear regions of ReLU networks affect their behavior and performance. Visual aids are provided to enhance reader understanding.

1. Introduction

Our main goal of this paper is to understand linear regions' impact on the performance and behavior in ReLU networks. The motivation behind exploring the linear regions of ReLU networks is to provide insights into how they function and perform. In a ReLU network, a linear region refers to the part of the input space where the activation function is linear. Linear regions can be determined by analyzing the boundaries between different regions in the input space. Linear regions in ReLU networks play a significant role in network expressivity, effectiveness, and information processing.

This paper aims at providing a comprehensive introduction to understanding the linear region of the ReLU networks. More precisely, Section 2 gives extra information to the reader to understand the terms and clear definitions of linear regions and how to count them theoretically and in practice. Then, Section 3 gives the definition of robustness and pieces of information about the relationship between linear regions and robustness because robustness is the ability to give good results even under bad conditions. Finally, we show some experiments on how linear regions affect the robustness of neural networks. After that in Section 4, we talk about

linear regions' effects on expressivity and performance. In this section, we talk about how linear regions make training faster and computationally optimal, and generalize better. Then, in section 5 we give the definition of what is shallow univariate ReLU network. After that, we give some articles' solutions and we try to point out what could be bad in real worlds solutions with examples and reasons. In section 6, we talk about how over-parameterization can benefit the ReLU networks. Finally, in section 7 we talk about our future plans and research about understanding linear regions.

Our main contributions are:

Firstly, giving a contra-argument about how deep and multi-variate ReLU networks can't give the same results as shallow and univariate ReLU networks. Finally, we give arguments about how over-parameterization can benefit the ReLU networks.

2. Linear Regions: Background

The concept of linear regions in ReLU networks can be elaborated by emphasizing that there are regions where the function is linear and the network behavior is simpler. Neurons in the first layer define linear boundaries that divide the input space into two regions, which are further combined and divided in subsequent layers, resulting in a network representation by different linear functions. Hanin, B. and Rolnik, D.(2019) give the equation for linear regions:

$\mathcal{B}_{\mathcal{N}}(\theta) := \{x \in \mathbb{R}^{(n_{in})} \mid \nabla \mathcal{N}(\cdot; \theta) \text{ is discontinuous at } x\}$
where θ a vector of trainable parameters for $\mathcal{N}$, and let $\mathcal{N}$ be a ReLU network with input n_{in} dimension

The linear regions of $\mathcal{N}$ at θ are the connected components of input space with $\mathcal{B}_{\mathcal{N}}$ removed:

$$\begin{aligned} & \text{linear regions}(\mathcal{N}, \theta) \\ &= \\ & \text{connected components}(\mathbb{R}^{(n_{in})} / \mathcal{B}_{\mathcal{N}}(\theta)) \end{aligned}$$

(1) This equation gives linear regions and there can be some differences when calculating activation regions.

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You can see there an example:

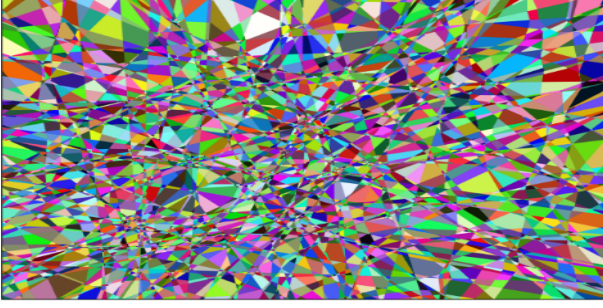


Figure: Hanin, B. and Rolnik, D.(2019) in this image show that a fully-connected ReLU network with three hidden layers of width 64 each is used in this image to represent a two-dimensional slice through the vectorized MNIST input space's 784 dimensions. The piecewise linear network's many linear areas are indicated by distinct colors.

Additionally, scholars attempt to determine the count of linear sections by theoretically predicting the bound, particularly the upper bound. With the present approaches, the upper bound is estimated throughout the whole input space. To estimate the number of linear regions within a certain region of the input space, such as a sphere centered around a training data point like an adversarial example or a backdoor trigger, theoretical approaches are currently needed. (Zhu et al., 2020)

On the other hand, there is the activation pattern which is very related to the linear regions. The activation pattern in a ReLU network can provide insights into how the network is processing the input data. It can help identify which neurons are active and firing, which can be useful for interpreting the network's behavior, debugging, or understanding the learned representations. Also, Hanin, B. and Rolnik, D.(2019) show an algorithm to count the activation patterns that can be used to count linear regions. This is the algorithm:

Let $\mathcal{N}$ be a feed-forward ReLU network with output dimension 1, no linked weights, input dimension n_{in} , and random weights/biases that satisfy:

1. There is a density in the distribution of all weights with regard to the Lebesgue measure on $\mathbb{R}^{\#weights}$.
2. Depending on the values of all weights and other biases, every collection of biases has a density with respect to the Lebesgue measure.
3. Because $C_{grad} > 0$ exists, for each neuron z and each $m \geq 1$, we have:

$$\sup_{x \in \mathbb{R}^{(n_{in})}} \mathbb{E}[(\|\nabla z(x)\|)^m] \leq (C_{grad})^m.$$

4. Given all the other weights and biases in $\mathcal{N}$, there exists

$C_{bias} > 0$ such that for any neurons $z_1, \dots, z_k$, the conditional distribution of the biases $p_{b_{z_1}, \dots, b_{z_k}}$ of these neurons fulfills:

$$\sup_{b_1, \dots, b_k \in \mathbb{R}} p_{b_{z_1}, \dots, b_{z_k}}(b_1, \dots, b_k) \leq (C_{bias})^k.$$

Then, there exists $\delta_0, T > 0$ depending on C_{grad}, C_{bias} with the following property. Suppose that $\delta > \delta_0$. Then, for all cubes C with side length δ , we have:

$$\frac{\mathbb{E}[\#non-empty activation regions of \mathcal{N} in C]}{vol(C)}$$

$\leq$

$$\begin{cases} (T \#neurons)^{n_{in}} / n_{in}!, & \#neurons \geq n_{in} \\ 2^{\#neurons}, & \#neurons \leq n_{in} \end{cases}$$

The average in this case refers to the weights and biases distribution in $\mathcal{N}$.

3. Relation between Linear Regions and Robustness

Robustness in neural networks refers to their ability to maintain accurate performance despite perturbations in the input data. In order to provide accurate predictions in real-world situations, it involves robustness against adversarial attacks, input noise, and variability. Robust neural networks withstand biases and weaknesses, and they generalize effectively. Building reliable and trustworthy AI systems depends on achieving robustness.

Robustness and linear regions are closely linked in ReLU networks. Linear regions are input regions where the network behaves linearly. Maximizing linear regions can enhance robustness by making the network less sensitive to small perturbations. Adversarial attacks exploit non-linearities, but having more linear regions makes it harder for attackers to find effective perturbations.

Croce et al. (2019) find a way to obtain either the ideal minimal perturbation or a lower constraint utilizing the characteristics of the linear region in which the point lies by taking use of the fact that these networks produce continuous piecewise affine classifier functions. By using that way, Croce et al. (2019) proposed a technique called Maximization of Linear Regions (MLR) that optimizes the network to have more linear regions. It is a technique that the regularization scheme for ReLU networks improves upon existing methods for adversarial training in terms of both lower and upper bounds on robustness. It can obtain classifiers that simultaneously provide strong robustness guarantees and good test error performance. However, achieving complete robustness requires considering other factors and employing multiple techniques. In addition, their experiments consider models trained to be L2-robust

and evaluate robustness as the average L2-norm of the perturbations necessary to change the classification when applied to test inputs. So, we can see the effectiveness of the proposed regularization scheme for ReLU networks.

L_2 -norm robustness						
Training Scheme	FC1			CNN		
	TE	LB	UB	TE	LB	UB
MNIST						
plain	1.73	9.7	66.3	0.85	3.1	100
at	1.15	2.6	16.9	0.87	1.8	100
KW	1.19	2.4	5.2	1.11	2.2	6.0
Xiao et al	-	-	-	-	-	-
MMR	2.40	5.9	8.8	2.57	5.8	11.6
MMR+at	1.77	3.8	6.4	2.12	4.6	9.7
CIFAR-10						
plain	-	-	-	23.31	47.2	100
at	-	-	-	25.82	35.8	100
KW	-	-	-	40.24	43.9	49.0
Xiao et al	-	-	-	-	-	-
MMR	-	-	-	40.92	50.6	57.1
MMR+at	-	-	-	37.75	43.9	53.3

Table: The statistical results of 5 distinct training approaches: plain (ordinary training), at (adversarial training), MMR (Croce et al. (2019)’s training style), MMR+at (MMR plus adversarial training), and KW (the robust training). For each dataset, it has been displaying the test error (TE), lower (LB), and upper (UB) bounds on the robust test error at that threshold.

4. Linear Regions Effects on Expressivity and Performance

The combination of linear and non-linear behavior allows ReLU networks to capture a wide range of patterns in the data. Linear regions contribute to expressivity by enabling the network to model simple linear relationships, while non-linear activations handle more complex patterns. Additionally, the linear behavior of ReLU activations simplifies computations, resulting in faster training and inference times, making ReLU networks computationally efficient. According to Hanin, B., and Rolnick, D. (2019), the linear regions also act as a form of regularization, helping to mitigate overfitting and improve generalization performance. However, it is important to note that an over-reliance on linear regions can limit the network’s ability to capture highly non-linear relationships, potentially leading to underfitting for complex tasks. Therefore, finding an appropriate balance between linear and non-linear behavior is crucial to achieving optimal performance in ReLU networks.

According to Hanin, B., and Rolnick, D. (2019), while training on a straightforward task does not result in a

significant rise in the number of regions, such tests leave open the possibility that if the task were more complicated, the number of regions would increase significantly. If the job is created so that having more regions is favorable, then the number of regions grows to achieve the theoretical upper bound (exponential in the depth). Memorization tasks are one apparent example because they call for learning highly oscillatory functions with numerous activation zones when applied to huge datasets. Also, you can see here their experiment result in memorization task:

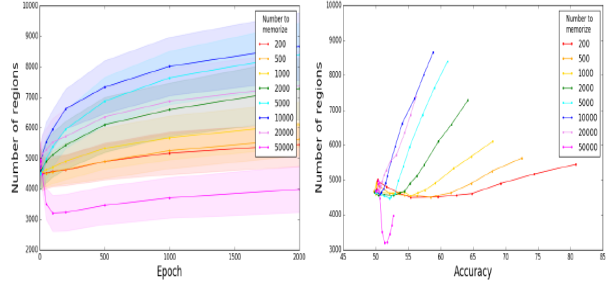


Figure: Depth 3, width 32, totally connected ReLU net was taught to memorize random 2D points with binary labels over 2000 epochs. Left: Plot illustrating the number of regions against period. Standard deviations are displayed with curves that represent the average of 40 separate training runs. Right: A plot of # regions versus training accuracy. During training, the number of regions grew, and it grew further for additional memorizing.

In addition, according to Hanin, B., and Rolnick, D. (2019), activation patterns that one measure of expressivity in ReLU networks and it grows with the depth of the ReLU network but its expressivity is bounded by total number of neurons raised to the input dimension.

Also, you can see the scopes here:

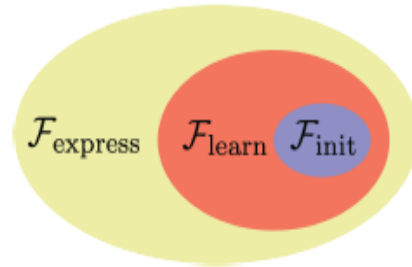


Figure: In this diagram, there is a set $\mathcal{F}_{express}$ of functions that can be expressed by a certain neural network design. The functions that correspond to networks at initialization are contained into a set called $\mathcal{F}_{init}$ within this set. A set

called $\mathcal{F}_{learn}$, which contains the functions that the network has a non-vanishing probability of learning using gradient descent, sits between $\mathcal{F}_{init}$ and $\mathcal{F}_{express}$. So, we can see difference between theoretical expressivity and neural network can actually learn in this diagram.

5. Expanding the Scope of Analysis with Shallow Univariate ReLU Networks

A neural network architecture made up of a single layer of ReLU activation functions is known as a shallow univariate ReLU network. In this usage, "shallow" refers to the network's few, often single, layers. The word "univariate" denotes that only one input variable or characteristic is used by the network to operate. Depending on the particular issue at hand, the network's output is generally employed for tasks like classification or regression.

Safran et al. (2023) show that gradient flow converges to a network achieving perfect training accuracy and having at most $O(r)$ linear regions, where r is the number of neurons in the shallow univariate ReLU network. This suggests a generalization bound and shows that optimization eventually leads to a solution that, up to a constant factor, actually possesses the ideal number of neurons.

Safran et al. (2023) point out that experiments that have been done can be done in multivariate ReLU networks but further research is needed both theoretically and practically. However, in contrast to a multivariate domain with neither output layer weights nor biases, Safran et al. (2023)'s experiments included a univariate domain with biases. This emphasizes how different the two architectures are, and it may not be possible to directly compare the constraints in the paper's results to those of multivariate ReLU networks with more hidden layers. Other than that, there are no known guarantees on the implicit bias for depth-2 ReLU networks trained with the square loss and the study of the implicit bias in the paper depends on particular results for exponentially-tailed classification losses.

In addition, we should consider some big differences between shallow and deep neural networks. While linear regions offer insights into network capacity, a comprehensive understanding of shallow ReLU networks requires considering broader dynamics. Focusing solely on linear regions oversimplifies network behavior, ignoring depth, architecture, and generalization. The practical relevance of theoretical guarantees is limited, as they may not hold in real-world scenarios. Non-linear interactions, specialized architectures, and empirical validation play crucial roles beyond linear regions. Bridging the gap between theoretical guarantees and practical performance is essential. Embracing a comprehensive analysis uncovers the true implications of ReLU networks beyond linear regions.

6. Over-Parameterization's effects on Linear Region

We investigate how shallow ReLU neural networks' optimal landscape is affected by slight over-parameterization. Before giving the reel object, we need to understand the teacher-student situation which is a well-researched situation in which the target values are produced by a teacher network made up of a sum of k neurons and taught by a student network of the same architecture with n neurons using the squared loss with respect to some input distribution D .

Safran et al. (2021) proved that the objective is not even locally convex after any amount of over-parameterization, despite being substantially convex around the global minima when the teacher and student networks have the same number of neurons.

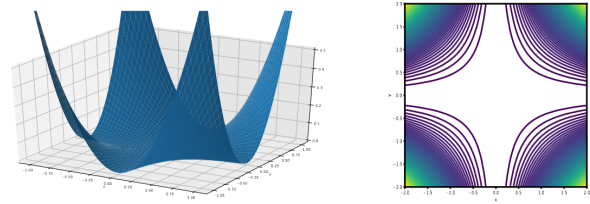


Figure: Plot and contour of the $f(x, y) = x^2 * y^2$ function. This function has a manifold of global minima with the values $x = 0$ or $y = 0$ and is continuously differentiable. Nearly everywhere that the determinant of its Hessian is negative, $f(x, y)$ is not locally convex. Additionally, it is not one-point strongly convex with respect to any global minimum.

These results imply that the optimal landscape of ReLU networks is significantly shaped by over-parameterization. By adding more neurons, the network becomes more flexible and has the ability to consider more options for solving problems. This improved adaptability enables local search techniques like gradient descent to move away from non-global minima and toward superior solutions. As a result, the optimization guarantee offered by this study lends credence to the idea that over-parameterization can improve the optimization process and increase neural networks' capacity for discovering global minima.

7. Future Research

To conclude, we concentrate on comprehending the function of linear regions of the ReLU networks and their influence on network expressivity, efficiency, and information processing. We go over the advantages and disadvantages of linear areas as well as how crucial it is to strike the right balance between linear and non-linear behavior in ReLU networks. Overall, the research offers insightful information about the operation of ReLU networks and their prospective use in deep learning.

In future studies, it will be important to explore the architectural design space of linear ReLU networks, investigate depth-width trade-offs and architectural variations because it affects how linear regions look, develop specialized regularization techniques, enhance interpretability and explainability of learned representations, and improve scalability and efficiency for large-scale datasets and complex architectures. These research directions hold the potential to advance our understanding and optimization of linear ReLU networks, leading to more effective and interpretable deep learning models in the future.

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